

Advanced numerical methods

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Orthogonal transformations I.

Orthogonal vectors

- We say that two vectors $\mathbf{u}, \mathbf{v} \in \mathcal{R}^n \setminus \{\mathbf{0}\}$ are *orthogonal*, if their inner product is zero:

$$\langle \mathbf{u}, \mathbf{v} \rangle \stackrel{def}{=} \mathbf{v}^T \mathbf{u} = 0$$

- Set of vectors is called orthogonal if its elements are pairwise orthogonal.
- The orthogonal vector-set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is *orthonormal*, if

$$\|\mathbf{u}_i\|_2 = 1 \quad (i = 1, \dots, k).$$

- The orthogonal and orthonormal vector-sets are two special cases of linearly independent vectors.

Orthogonal matrices

- The matrix $\mathbf{Q} \in \mathcal{R}^{n \times n}$ is called *orthogonal*, if its column vectors form an orthonormal vector-set.
- In this case it holds for matrix $\mathbf{Q}$ that

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{Q} \mathbf{Q}^T = \mathbf{I}$$

- From this equation it immediately follows that

$$\mathbf{Q}^T = \mathbf{Q}^{-1}$$

meaning that inverting an orthogonal matrix is easy.

- Let's see the properties of orthogonal matrices.

2.1. Lemma. *Multiplication of orthogonal matrices is orthogonal.*

2.2. Lemma. *The matrix $\mathbf{Q} \in \mathcal{R}^{n \times n}$ is orthogonal if and only if $\|\mathbf{x}\|_2 = \|\mathbf{Q}\mathbf{x}\|_2$ holds for any vector $\mathbf{x} \in \mathcal{R}^n$.*

2.3. Lemma. *For any matrix $\mathbf{A} \in \mathcal{R}^{n \times n}$ and orthogonal matrix $\mathbf{Q}$ we have*

(i) $\|\mathbf{QA}\|_2 = \|\mathbf{A}\|_2$, and

(ii) $\|\mathbf{QA}\|_F = \|\mathbf{A}\|_F$.

2.4. Corollary. *For any matrix $\mathbf{A} \in \mathcal{R}^{n \times n}$ and orthogonal matrix $\mathbf{Q}$ we have*

(i) $\text{cond}(\mathbf{QA}) = \text{cond}(\mathbf{A})$ and

(ii) $\text{cond}(\mathbf{Q}^T \mathbf{A} \mathbf{Q}) = \text{cond}(\mathbf{A})$.

Definition of condition number: $\text{cond}(\mathbf{A}) = \|\mathbf{A}^{-1}\| \|\mathbf{A}\|$

Rule of thumb: huge condition number means instability during solving a system of linear equations.

Reflection matrices

(Householder matrix)

Elementary reflector matrix

- In the followings we are looking for the decomposition of matrix $\mathbf{A}$ in the form

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{Q}$ is orthogonal, $\mathbf{R}$ is upper-triangular matrices.

- A procedure was given by Alston Householder in 1958, which we can call *orthogonal triangularization*.
- In this procedure we apply elementary matrix operations using reflectors in $\mathcal{R}^n$.
- we will be looking for the matrix $\mathbf{R}$ and obtain matrix $\mathbf{Q}$ as side-result.

We are defining first the matrix form of reflexion (from elementary geometry).

Example

2.6. Example. Take the vectors $\mathbf{x} = (-1, 3)^T$ and $\mathbf{h} = (2, 3)^T$.

Now $\|\mathbf{h}\|_2 = \sqrt{13}$ and we obtain the reflector

$$\mathbf{H} = \begin{pmatrix} 5/13 & -12/13 \\ -12/13 & -5/13 \end{pmatrix}$$

The result of $\mathbf{H}\mathbf{x}$ is the vector $(-41/13, -3/13)^T$.

For another example (drawing) see the lecture notes.

$$(-3.153846154, -0.230769231)^T$$

Let's prove two properties of reflection matrices.

2.7. Lemma. *For all reflection matrix $\mathbf{H}$ the followings are true:*

(i) $\mathbf{H}^T = \mathbf{H}$ and

(ii) $\mathbf{H}^{-1} = \mathbf{H}$,

i.e., $\mathbf{H}$ is symmetric and orthogonal.

Proof. The case $\mathbf{H} = \mathbf{I}$ is trivial.

(i) Matrix $\mathbf{H}$ is indeed symmetric:

$$\mathbf{H}^T = \left(\mathbf{I} - \frac{2}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T \right)^T = \mathbf{I}^T - \frac{2}{\|\mathbf{h}\|_2^2} \left(\mathbf{h}^T \right)^T \mathbf{h}^T = \mathbf{I} - \frac{2}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T = \mathbf{H}.$$

(ii) To prove the orthogonality examine the product $\mathbf{H}^T \mathbf{H}$:

$$\begin{aligned}\mathbf{H}^T \mathbf{H} &= \mathbf{H} \mathbf{H} \\&= \left(\mathbf{I} - \frac{2}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T \right) \left(\mathbf{I} - \frac{2}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T \right) \\&= \mathbf{I} - \frac{2}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T - \frac{2}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T + \frac{4}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T \frac{1}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T \\&= \mathbf{I} - \frac{4}{\|\mathbf{h}\|_2^2} \mathbf{h} \mathbf{h}^T + 4 \mathbf{h} \underbrace{\left(\frac{\mathbf{h}^T \mathbf{h}}{\|\mathbf{h}\|_2^2} \right)}_1 \mathbf{h}^T \frac{1}{\|\mathbf{h}\|_2^2} \\&= \mathbf{I},\end{aligned}$$

which means that $\mathbf{H}$ is orthogonal. ■

In the next lemma we show that given two vectors of equal length, there is always an elementary reflection which transforms one into the other one.

2.8. Lemma. *Let $\mathbf{a}, \mathbf{b} \in \mathcal{R}^n$ be two distinct vectors, none of them equal to $\mathbf{0}$. If $\|\mathbf{a}\|_2 = \|\mathbf{b}\|_2$, then the reflector defined by*

$$\mathbf{h} \stackrel{\text{def}}{=} (\mathbf{a} - \mathbf{b}) \quad \text{and} \quad \mathbf{H} \stackrel{\text{def}}{=} \mathbf{I} - 2 \frac{(\mathbf{a} - \mathbf{b})(\mathbf{a} - \mathbf{b})^T}{\|\mathbf{a} - \mathbf{b}\|_2^2}$$

transforms $\mathbf{a}$ into $\mathbf{b}$.

Before we discuss the proof, note that

$$\|\mathbf{a}\|_2 = \|\mathbf{b}\|_2 \iff \mathbf{a}^T \mathbf{a} = \mathbf{b}^T \mathbf{b}$$

Proof. Calculate the vector $\mathbf{H}\mathbf{a}$:

$$\begin{aligned}\mathbf{H}\mathbf{a} &= \left(\mathbf{I} - 2 \frac{(\mathbf{a} - \mathbf{b})(\mathbf{a} - \mathbf{b})^T}{\|\mathbf{a} - \mathbf{b}\|_2^2} \right) \mathbf{a} \quad (\text{using } (a - b)^T a = a^T (a - b)) \\ &= \mathbf{a} - \frac{2\mathbf{a}^T(\mathbf{a} - \mathbf{b})}{(\mathbf{a} - \mathbf{b})^T(\mathbf{a} - \mathbf{b})}(\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a} - \frac{2(\mathbf{a}^T \mathbf{a} - \mathbf{a}^T \mathbf{b})}{\mathbf{a}^T \mathbf{a} - \mathbf{a}^T \mathbf{b} - \mathbf{b}^T \mathbf{a} + \mathbf{b}^T \mathbf{b}}(\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a} - \frac{2(\mathbf{a}^T \mathbf{a} - \mathbf{a}^T \mathbf{b})}{2(\mathbf{a}^T \mathbf{a} - \mathbf{a}^T \mathbf{b})}(\mathbf{a} - \mathbf{b}) \quad (\text{using } a^T a = b^T b, \text{ which was our assumption}) \\ &= \mathbf{b}.\end{aligned}$$

meaning that $\mathbf{H}$ is indeed the linear transformation we were looking for. ■

Application of the previous lemma

- Later we will be applying the previous transformation in the special case $\mathbf{b} = \pm \|\mathbf{a}\|_2 \mathbf{e}_1$
- here the aim of the reflection is to eliminate all components of vector $\mathbf{a}$, *except the first one*.
- Here we use the transformation given by the vector $\mathbf{h} = \mathbf{a} \pm \|\mathbf{a}\|_2 \mathbf{e}_1$.
- It is useful to choose the sign in a way that in the first component should be $h_1 = a_1 + \text{sgn}(a_1) \|\mathbf{a}\|_2$, which increases the stability of computations.

Illustration?

Complexity

2.9. Lemma. *Given an elementary reflection matrix $\mathbf{H}$, a matrix $\mathbf{A}$ and a vector $\mathbf{x}$ we can get*

- *vector $\mathbf{Hx}$ by $O(n)$ computations,*
- *matrix $\mathbf{HA}$ by $O(n^2)$ computations.*

Proof. Sketch: multiplication by matrix $\mathbf{H}$ is basically a multiplication by a matrix of form $\mathbf{S} = \mathbf{uv}^T$, where $\mathbf{u} = \mathbf{v}$. Thus we have

- $\mathbf{Sx} = (\mathbf{uv}^T)\mathbf{x} = \mathbf{u}(\mathbf{v}^T\mathbf{x}) = \mathbf{u}\beta = \beta\mathbf{u}$, where $\beta = \mathbf{v}^T\mathbf{x}$
- $\mathbf{SA} = (\mathbf{uv}^T)\mathbf{A} = \mathbf{u}(\mathbf{v}^T\mathbf{A}) = \mathbf{uz}^T$, where $\mathbf{z}^T = \mathbf{v}^T\mathbf{A}$.



Partitioned form

The Householder matrices can be also given in partitioned form.

2.10. Lemma. *Let $1 \leq k < n$ be arbitrary, and $\mathbf{H} \in \mathcal{R}^{k \times k}$ be arbitrary elementary reflector. Then the matrices $\mathbf{T}$ and $\mathbf{W}$ (both are n dimensional) defined by*

$$\mathbf{T} \stackrel{\text{def}}{=} \left(\begin{array}{c|c} \mathbf{I} & \mathbf{O} \\ \hline \mathbf{O}^T & \mathbf{H} \end{array} \right) \in \mathcal{R}^{n \times n} \quad \text{and} \quad \mathbf{W} \stackrel{\text{def}}{=} \left(\begin{array}{c|c} \mathbf{H} & \mathbf{O} \\ \hline \mathbf{O}^T & \mathbf{I} \end{array} \right) \in \mathcal{R}^{n \times n}$$

will also be elementary reflection matrices.

Orthogonal triangularization by Householder method

We need to emphasize that the method can be applied on any matrix $\mathbf{A}$.

2.11. Theorem. *The orthogonal triangularization of any matrix $\mathbf{A} \in \mathcal{R}^{n \times n}$ can be given by finite number of multiplication of elementary reflection matrices. More precisely, we can define the series of elementary reflection matrices $\mathbf{H}_1, \mathbf{H}_2, \dots, \mathbf{H}_{n-1}$ such that the recursion*

$$\mathbf{A}_0 \stackrel{\text{def}}{=} \mathbf{A}, \quad \mathbf{A}_j \stackrel{\text{def}}{=} \mathbf{H}_j \mathbf{A}_{j-1} \quad j = 1, 2, \dots, n-1$$

results in a matrix $\mathbf{A}_j$ which is upper triangular in its columns $1, 2, \dots, j$. Specifically, the matrix $\mathbf{A}_{n-1}$ is upper triangular.

Proof.

- We prove by induction.
- In case of $j = 1$ we apply the previous lemma (see on the white-board) choosing
 - $\mathbf{a} = \mathbf{A} \mathbf{e}_1 = \mathbf{a}_1$
 - and $\mathbf{b} = \operatorname{sgn}(a_{11}) \|\mathbf{a}_1\|_2 \mathbf{e}_1$.
- Hence, the multiplication $\mathbf{H}_1 \mathbf{A}_0$ results in $\mathbf{A}_1$ which first column is upper triangular.

- Assume that the theorem is true for $i = 1, 2, \dots, j$.
- Write matrix $\mathbf{A}_j$ in the partitioned form

$$\mathbf{A}_j = \left(\begin{array}{c|c} \mathbf{R}_j & \mathbf{B}_j \\ \hline \mathbf{O}_j^T & \mathbf{C}_j \end{array} \right)$$

where $\mathbf{R}_j \in \mathcal{R}^{j \times j}$, $\mathbf{B}_j \in \mathcal{R}^{j \times (n-j)}$, $\mathbf{O}_j^T \in \mathcal{R}^{(n-j) \times j}$, and $\mathbf{C}_j \in \mathcal{R}^{(n-j) \times (n-j)}$.

- We know that matrix $\mathbf{R}_j$ is upper triangular (that is the assumption of the induction)
- we need to deal with the matrix $\mathbf{C}_j$ only.
- We apply the lemma again...

... we can find an elementary reflection matrix $\hat{\mathbf{H}}_{j+1} \in \mathcal{R}^{(n-j) \times (n-j)}$ with which if we multiply the matrix $\hat{\mathbf{H}}_{j+1} \mathbf{C}_j$ we get a matrix in which the first column has 0 elements from the second row and below.

We know that

$$\mathbf{H}_{j+1} = \left(\begin{array}{c|c} \mathbf{I}_j & \mathbf{O}_j \\ \hline \mathbf{O}_j^T & \hat{\mathbf{H}}_{j+1} \end{array} \right) \in \mathcal{R}^{n \times n}$$

is also elementary reflection matrix

$$\mathbf{A}_{j+1} = \mathbf{H}_{j+1} \mathbf{A}_j = \left(\begin{array}{c|c} \mathbf{I}_j & \mathbf{O}_j \\ \hline \mathbf{O}_j^T & \hat{\mathbf{H}}_{j+1} \end{array} \right) \left(\begin{array}{c|c} \mathbf{R}_j & \mathbf{B}_j \\ \hline \mathbf{O}_j^T & \mathbf{C}_j \end{array} \right) = \left(\begin{array}{c|c} \mathbf{R}_j & \mathbf{B}_j \\ \hline \mathbf{O}_j^T & \hat{\mathbf{H}}_{j+1} \mathbf{C}_j \end{array} \right)$$

which is upper triangular in its $j+1$ -th column, thus we proved the theorem for $j+1$.

■

- We can see that

$$\mathbf{H}_{n-1}\mathbf{H}_{n-2}\cdots\mathbf{H}_2\mathbf{H}_1\mathbf{A}_0 = \mathbf{A}_{n-1},$$

hence, by reordering we obtain

$$\mathbf{A}_0 = (\mathbf{H}_1\mathbf{H}_2\cdots\mathbf{H}_{n-2}\mathbf{H}_{n-1})\mathbf{A}_{n-1}, \quad (\text{how is that possible??})$$

thus, with the definitions of

$$\mathbf{Q} \stackrel{\text{def}}{=} \mathbf{H}_1\mathbf{H}_2\cdots\mathbf{H}_{n-2}\mathbf{H}_{n-1}$$

and

$$\mathbf{R} \stackrel{\text{def}}{=} \mathbf{A}_{n-1}$$

we indeed obtain the orthogonal triangular decomposition $\mathbf{A} = \mathbf{QR}$, where we used the fact in the definition of matrix $\mathbf{Q}$ that multiplication of orthogonal matrices results in orthogonal matrix.

Complexity

- By Lemma 2.9 the multiplication of a matrix $\mathbf{A}$ with a Householder matrix has complexity $O(n^2)$.
- The proof of Theorem 2.11 showed that we need $n - 1$ multiplications.
- Hence, the orthogonal triangularization $\mathbf{A} = \mathbf{QR}$ by Householder transformations needs $O(n^3)$.